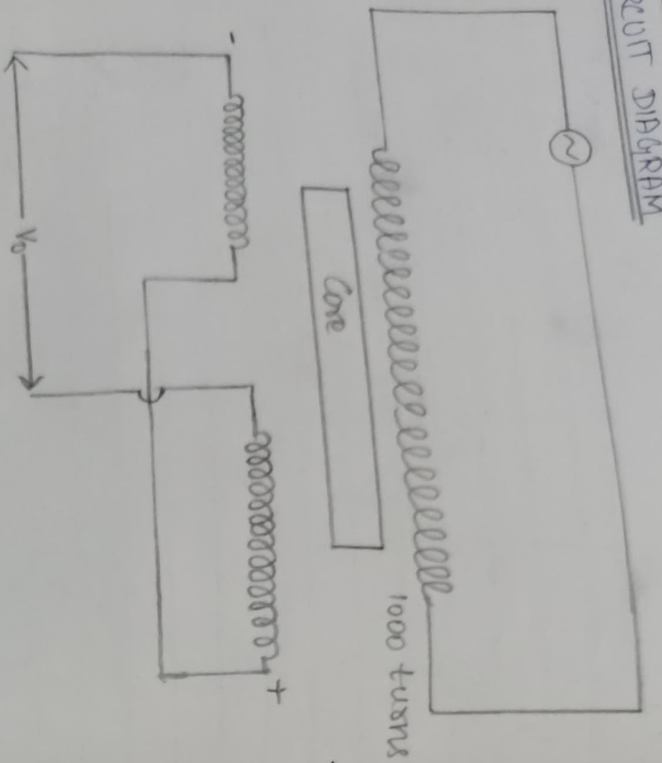


CIRCUIT DIAGRAM



TRBUATION:

Input voltage (V _{in})	Output voltage	Frequency (Hz)	Time = $\frac{1}{f}$
Positive displacement	2.2V (D=5mm)	1000	$\frac{1}{1000} = 1\text{ms}$
Negative displacement	(D=-5mm)	1000	1ms

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CHARACTERISTICS OF LVDT (LINEAR VARIABLE

DIFFERENTIAL TRANSFORMER)

AIM:

To understand and stimulate the relation between core displacement and output voltage.

APPARATUS REQUIRED:

Laptop with Internet connection

THEORY:

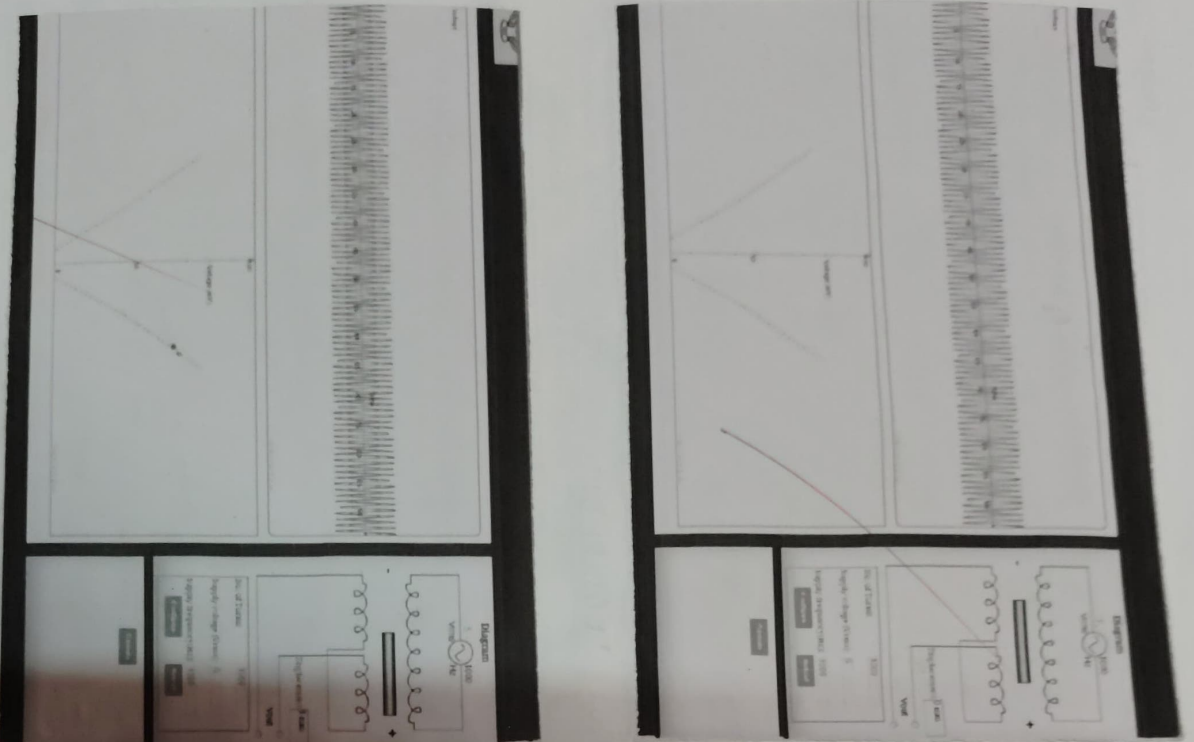
LVDT is an inductive transducer that converts linear displacement into an electrical signal. It consists of a transformer having one primary winding and two secondary windings wound on a core. As the primary is connected to an AC source, the AC current and the voltage are produced in the secondary of LVDT.

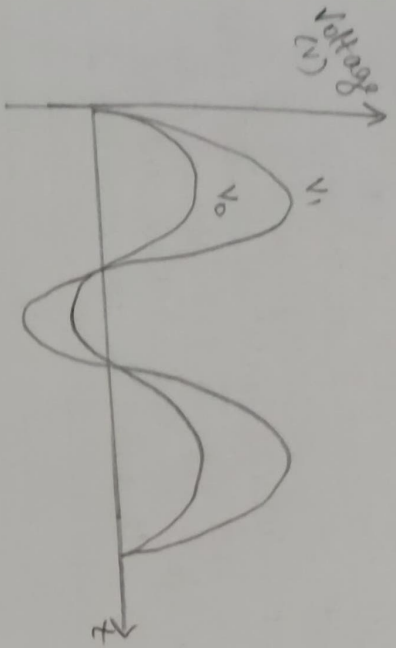
When the core is at the centre, the flux linking with both the secondary windings are equal, so the induced EMF in both the windings

are equal, this means there is a no displacement when the core is at the right, the flux linking with the secondary winding S_1 is more. So, the emf induced in S_1 is more than S_2 . So, the net emf is positive. When the core is at the left, the flux linking and the emf in S_2 is more than S_1 . So, the net emf is negative.

PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Set the number of turns supplied voltage, and supplied frequency.
3. Configure the parameters.
4. Move the core to positive side and the negative side and observe the input voltage and output waveforms.
5. Plot the graph between input and output voltage.



POSITIVE DISPLACEMENTCalculation

$$V_m = \sqrt{2} \times V_{rms}$$

$$V_m = \sqrt{2} \times 5$$

$$V_m = 7V$$

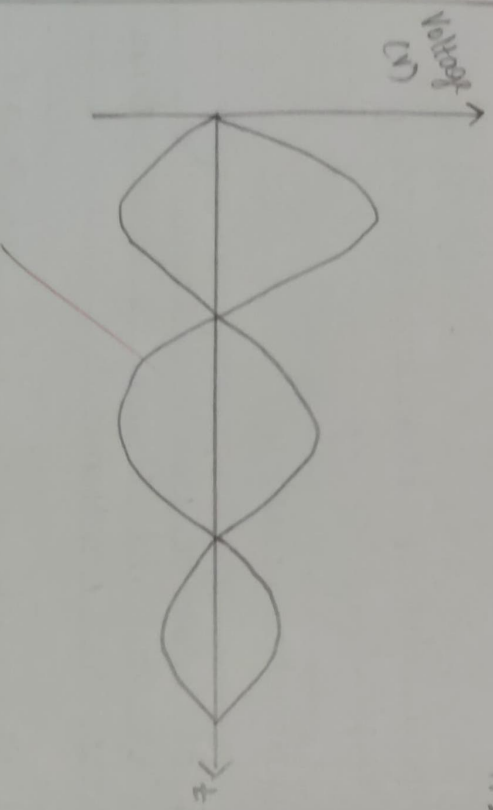
$$T_1 = 1ms$$

$$4SD = 7$$

$$1SD = 7/4 V$$

$$1.25SD = 7/4 \times 1.25$$

$$1.25SD = 2.2V$$

NEGATIVE DISPLACEMENTRESULT:

To understand and simulate the relation between core displacement and output voltage were developed and tested successfully.

18/08/21

